Lab 2 - Product Specification

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Table of Contents

1.1 Purpose.....3
1.2 Scope.....3
1.3 Definitions, Acronyms, and Abbreviations..... 5
1.4 References.....7
1.5 Overview.....8
2.0 Overall Description.....9
2.1 Product Perspective.....9
2.2 Product Functions..... 10
2.3 User Characteristics..... 12
2.4 Constraints..... 12
2.5 Assumptions and Dependencies..... 12

Table of Figures

Figure 1: Major Functional Components Diagram (MFCD).....9
Figure 2: RWP vs Prototype vs Current Plan Table.....11

1.1 Purpose

The purpose of this document is to have a clearly defined reference for developers and stakeholders for implementation of the system. This is a Software Requirements Specification (SRS). Developers refer to this document for a clear understanding of what is expected to be implemented. This can also be a vital tool for developers to create user stories for focused and productive development. Testers also benefit from clear definitions of expectations listed within the document. As developers create user stories for implementation, testers can use those user stories to make sure implementation is accurate. Lastly, stakeholders set the expectations for the developer to follow, everything laid out should be specified and explained elaborately so everyone's needs are met. Features displayed by the system, constraints, and assumptions must all be accounted for and included within this document.

This document contains information about Team Copper's web-application AI^2. Details about the scope of this application, purpose, functionality, constraints and definitions are listed ahead.

1.2 Scope

The goal of AI^2 is to streamline invoice management and matching through automation. AP team members previously responsible for matching can focus on unique cases where invoices and POs do not match or information is missing.

The system will automate invoice intake upon arrival and be sorted. The system will match invoices based on their corresponding purchase order. The system will provide a confidence score based on the certainty that the correct purchase order to invoice matches. The

system will notify a human for review if a PO match has a low confidence score. The system will use user feedback to improve matching between invoices and PO's. The system will provide audit logging to track decisions for review if necessary.

The system will not instill a prioritization system for emails that are not invoice-related. The system will not replace the user within the invoice to PO matching workflow.

1.3 Definitions, Acronyms, and Abbreviations

AI² (Artificially Intelligent Invoices): The proposed automated invoice processing system designed to streamline Accounts Payable workflows through AI-driven matching, classification, and routing.

Alteryx: A data automation and analytics tool that enables secure API connections and data workflows between applications.

AP (Accounts Payable): The amount a business owes for goods and services purchased on credit; typically due at intervals of 30, 45, 60, or 90 days depending on repayment terms.

APEX: A strongly typed, object-oriented programming language that Salesforce developers use to execute flow and transaction control statements on the Salesforce platform.

AWS GovCloud: A secure U.S. government-compliant cloud environment that supports sensitive data and regulated workloads in accordance with FedRAMP and DoD requirements.

Atlantic Diving Supply (ADS): An American federal contractor company that provides equipment and logistics solutions to the Department of Defense, federal agencies, and first responders.

CMMC (Cybersecurity Maturity Model Certification): A framework developed by the U.S. Department of Defense to assess and enhance the cybersecurity posture of contractors handling federal information.

ERP (Enterprise Resource Planning System): An integrated software system used to manage core business processes, such as finance, procurement, and supply chain operations.

FedRAMP (Federal Risk and Authorization Management Program): A U.S. government program that standardizes security assessment, authorization, and continuous monitoring for cloud services.

Graph API: A Microsoft API interface used to access and automate data operations within Outlook and other Microsoft 365 services.

Invoice: A document listing goods or services provided, including a statement of the amount due; a bill.

Oracle: The system of record used by ADS for enterprise data and financial management.

PO (Purchase Order): The official order list ADS sends to suppliers detailing the items or services requested.

Salesforce: A robust Customer Relationship Management (CRM) platform that allows businesses to manage customer relationships efficiently.

Supplier: A person or organization that provides goods or services needed by another organization.

Vendor: A person or company offering goods or services for sale.

1.4 References

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1.5 Overview

The following sections of this document include product details, key features implemented/eliminated, user roles, and constraints. Section 2 provides an overall description of the system including product perspective, functions, user characteristics, constraints, and dependencies. To go in more detail, Section 2.1 summarizes the purpose and reasoning behind developing AI^2. Section 2.2 lists out functions that are to be implemented into the project, and how significant they are to how the application works as a whole. The user characteristics Section 2.3 defines the different types of user roles that the application takes account for. Lastly Sections 2.4 and 2.5 details the risks of applying this application to a work environment and dependencies on a third-party resource, respectively.

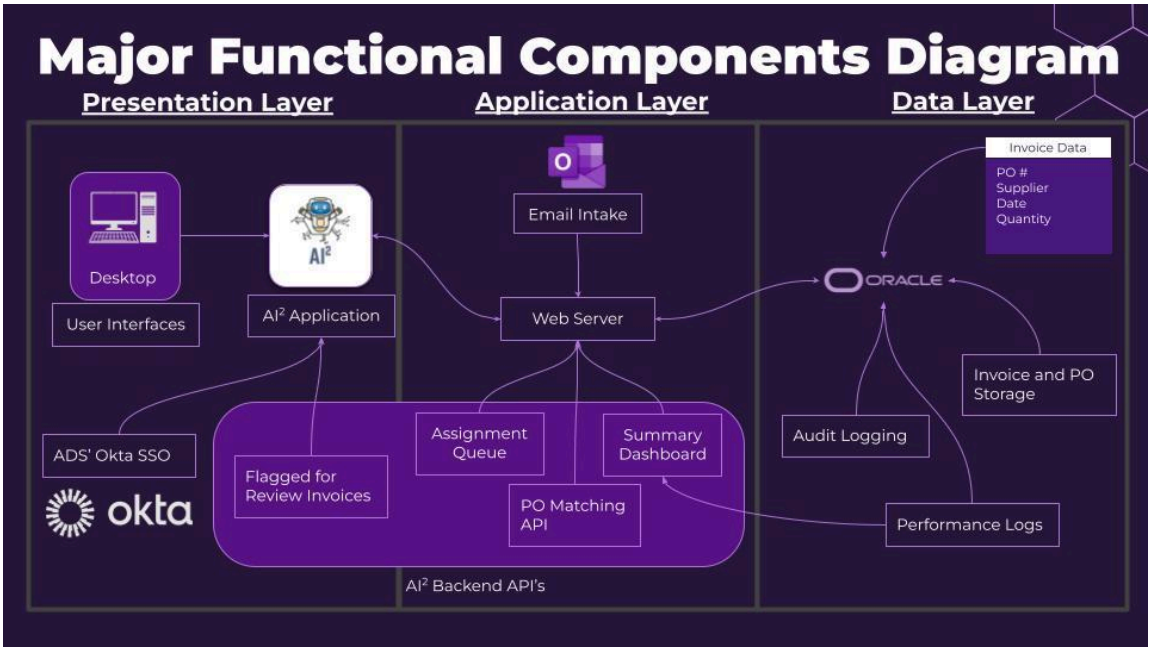
Section 3 will detail the specific software requirements.

2.0 Overall Description

2.1 Product Perspective

Team Copper is building AI², an invoice management web-application for automated invoice intake, information parsing, and matching to a corresponding purchase order. One of the key features of AI² is the ability for human reviewers to correct and improve the application's matching algorithm through feedback. The goal of AI² is not to replace human reviewers, but to leave their skills and time to more unique cases where invoices and purchase orders do not match.

Figure 1: Major Functional Components Diagram (MFCD)



The application starts with the presentation layer authenticating users through Okta SSO. AP team members will then be interacting with the assignment queue where invoices are flagged for human review. The application layer contains the invoice intake through Microsoft Outlook

and parsing through automated processes. The invoice to purchase order matching is also done within this layer. The data layer will store matched invoices and purchase orders.

2.2 Product Functions

The features that will be fully implemented are the invoice parser, data extractor, and the invoice quality analyzer. The invoice parser is essential to AI^2 as without it there is no matching the purchase orders. The invoice quality analyzer is just as integral, as it will provide AI^2 the ability to identify unique cases that should be left for human review. If invoices are received with missing or difficult to identify information this feature will highlight that to the user.

Partially implemented features include: automated PO matching algorithm, confidence scoring module, continuous learning from human interaction. Automated PO matching, cannot be fully implemented due to how versatile this feature could be. We are entirely focused on a specific format of invoice/PO (pdf files) and how different they look that currently we cannot expand our scope to account for every file type and its format. The confidence scoring feature is simply displayed with either a 1 or a 0, 1 meaning it has a match and 0 meaning no match, thresholds for matching have yet to be fully determined.

Features that are entirely eliminated from the project are implementations that are 3rd party such as Microsoft Outlook, Okta SSO, AWS GovCloud, etc.

[Space intentionally left blank, RWP vs Prototype vs Current Plan Table below]

Figure 2: RWP vs Prototype vs Current Plan Table

RWP vs Prototype vs Current Plan

Feature	Real World Product	Prototype	Current Plan
Outlook Email Integration (Graph API)	Fully Functional	Eliminated	Eliminated
Invoice Parser and Data Extractor	Fully Functional	Fully Functional	Fully Functional
Automated PO Matching Algorithm	Fully Functional	Partially Functional	Fully Functional
Invoice Quality Analyzer	Fully Functional	Not Implemented	Fully Functional
Queue Assigning and Routing	Fully Functional	Not Implemented	Partially Functional
Queue Assignment Dashboard	Fully Functional	Not Implemented	Partially Functional

22

RWP vs Prototype vs Current Plan

Feature	Real World Product	Prototype	Current Plan
Model Training and Deployment Pipeline	Fully Functional	Not Implemented	Fully Functional
Model Version Control	Fully Functional	Not Implemented	Fully Functional
AI Performance Dashboard	Fully Functional	Not Implemented	Fully Functional
Invoice Data Anonymization Pipeline	Fully Functional	Eliminated	Eliminated

23

RWP vs Prototype vs Current Plan

Feature	Real World Product	Prototype	Current Plan
Admin Configuration Dashboard	Fully Functional	Not Implemented	Partially Functional
System Health and Performance Dashboard	Fully Functional	Not Implemented	Partially Functional
Error Logging and Recovery Module	Fully Functional	Not Implemented	Fully Functional
Admin System Alert Notifications	Fully Functional	Not Implemented	Partially Functional

24

RWP vs Prototype vs Current Plan

Feature	Real World Product	Prototype	Current Plan
Automated Supplier Notifications	Fully Functional	Eliminated	Eliminated
Automated AP Team Reminders	Fully Functional	Not Implemented	Partially Functional
AI Confidence Scoring Module	Fully Functional	Not Implemented	Fully Functional
Continuous Learning From Human Interaction	Fully Functional	Not Implemented	Fully Functional
Human Review Panel	Fully Functional	Not Implemented	Fully Functional

25

RWP vs Prototype vs Current Plan

Feature	Real World Product	Prototype	Current Plan
Encryption Verification Tool	Partially Functional	Eliminated	Eliminated
AWS GovCloud Integration Tool	Fully Functional	Eliminated	Eliminated
Audit Log Management System	Fully Functional	Not Implemented	Partially Functional
Compliance Changelog System	Fully Functional	Not Implemented	Partially Functional
Auto Generated Compliance Report	Fully Functional	Eliminated	Eliminated
Compliance Regulations Monitoring Tool	Fully Functional	Eliminated	Eliminated

26

RWP vs Prototype vs Current Plan

Feature	Real World Product	Prototype	Current Plan
Okta SSO Authentication Module	Fully Functional	Eliminated	Eliminated
Alteryx-Oracle Integration Module	Fully Functional	Eliminated	Eliminated
Integration Monitoring Dashboard	Fully Functional	Eliminated	Eliminated

27

2.3 User Characteristics

The Administrator role is responsible for overseeing the AI^2 process. They are responsible for collaborating with the AP team members to distribute the workload if it gets overwhelming for someone. Confidence scoring thresholds are capable of being adjusted through the administrator role.

The AP Team member role is responsible for dealing with invoices and purchase orders that do not match. On the other hand, they are also reviewing and correcting AI^2 matches based on the confidence score. AI^2 is built around supporting the AP team members as much as possible.

2.4 Constraints

Incorrect automation results could lead to financial loss if invoice-to-PO matches are inaccurate. OCR and parsing errors may occur due to automation of invoice processing. Accuracy for edge cases will be low, but should be an expected outcome. Excessive manual corrections are likely when automating complex processes. The system handles sensitive financial client data, requiring appropriate security measures.

2.5 Assumptions and Dependencies

AI2 depends on Okta SSO for user authentication. Microsoft Outlook for automated invoice intake via the Graph API, and Oracle as the ERP system of record for purchase order data.